

### Advanced (Habanero)

- Q1.** Given  $f(x) = x^2$  and  $g(x) = 2x$ , what is  $f(x) + g(x)$ ?  
(A)  $x^2 + 2x$   
(B)  $x^2 - 2x$   
(C)  $2x^2$   
(D)  $x^2 + x$   
(E)  $2x^2 + x$
- Q2.** What is the domain of  $f(x) = \frac{1}{x}$  and  $g(x) = \frac{1}{x+1}$  when combined?  
(A)  $(-\infty, -1) \cup (-1, \infty)$   
(B)  $(-\infty, -1) \cup (-1, 0) \cup (0, \infty)$   
(C)  $(-\infty, 0) \cup (0, \infty)$   
(D)  $(-\infty, -1) \cup (0, \infty)$   
(E)  $(-1, 0) \cup (0, \infty)$
- Q3.** If  $f(x) = x^2 - 4$  and  $g(x) = x + 1$ , what is  $f(x) - g(x)$ ?  
(A)  $x^2 - x - 5$   
(B)  $x^2 - x - 3$   
(C)  $x^2 + x - 3$   
(D)  $x^2 - 5$   
(E)  $x^2 + x + 3$
- Q4.** Given  $f(x) = 2x$  and  $g(x) = x - 1$ , what is  $f(x) \cdot g(x)$ ?  
(A)  $2x^2 - 2x$   
(B)  $2x^2 + 2x$   
(C)  $x^2 - 2x$   
(D)  $2x^2 - x$   
(E)  $x^2 + x$
- Q5.** What is the simplified form of  $\frac{f(x)}{g(x)}$  when  $f(x) = x^2$  and  $g(x) = x$ ?  
(A)  $x$   
(B)  $x^2$   
(C)  $\frac{1}{x}$   
(D)  $\frac{1}{x^2}$   
(E)  $x^3$
- Q6.** Given  $f(x) = x + 1$  and  $g(x) = x - 1$ , what is  $f(x) + g(x)$ ?  
(A)  $2x$   
(B)  $2x + 1$   
(C)  $2x - 1$   
(D)  $x + 2$   
(E)  $x - 2$
- Q7.** If  $f(x) = \frac{1}{x}$  and  $g(x) = x$ , what is  $f(x) \cdot g(x)$ ?  
(A) 1  
(B)  $x$   
(C)  $x^2$   
(D)  $\frac{1}{x}$   
(E)  $\frac{1}{x^2}$
- Q8.** What is the domain of  $f(x) = \frac{1}{x-1}$  and  $g(x) = \frac{1}{x+2}$  when combined?  
(A)  $(-\infty, -2) \cup (-2, 1) \cup (1, \infty)$   
(B)  $(-\infty, -2) \cup (1, \infty)$   
(C)  $(-\infty, 1) \cup (1, \infty)$   
(D)  $(-2, 1) \cup (1, \infty)$   
(E)  $(-\infty, -2) \cup (-2, 1)$

#### Open Ended Questions (FRQ)

- Q9.** Given  $f(x) = x^2 + 2x$  and  $g(x) = x - 2$ , find  $f(x) \cdot g(x)$  and simplify the expression. Show all your work.
- Q10.** If  $f(x) = \frac{1}{x+1}$  and  $g(x) = x^2$ , determine the domain of  $f(x) + g(x)$ . Provide a clear explanation for your answer.

#### Chef's Tips

- Ensure students show all work for free response questions
- Remind students to simplify expressions whenever possible
- Check for understanding of domain restrictions in combined functions